TopoLogic: An Interpretable Pipeline for Lane Topology Reasoning on Driving Scenes

Yanping Fu
Institute of Computing Technology,
Chinese Academy of Sciences
University of Chinese Academy of Sciences
fuyanping23s@ict.ac.cn

Wenbin Liao
Institute of Computing Technology,
Chinese Academy of Sciences
University of Chinese Academy of Sciences
liaowenbin23z@ict.ac.cn

Xinyuan Liu
Institute of Computing Technology,
Chinese Academy of Sciences
University of Chinese Academy of Sciences
liuxinyuan21s@ict.ac.cn

Hang Xu
Hangzhou Dianzi University
hxu@hdu.edu.cn

Yike Ma
Institute of Computing Technology,
Chinese Academy of Sciences
ykma@ict.ac.cn

Yucheng Zhang
Institute of Computing Technology,
Chinese Academy of Sciences
zhangyucheng@ict.ac.cn

Feng Dai*
Institute of Computing Technology,
Chinese Academy of Sciences
fdai@ict.ac.cn

Abstract

As an emerging task that integrates perception and reasoning, topology reasoning in autonomous driving scenes has recently garnered widespread attention. However, existing works often emphasizes "perception over reasoning": they typically boost reasoning performance by enhancing the perception of lanes and directly adopt vanilla MLP to learn lane topology from lane query. This paradigm overlooks the geometric features intrinsic to the lanes themselves and is prone to being influenced by inherent endpoint shifts in lane detection. To tackle this issue, we propose an interpretable method for lane topology reasoning based on lane geometric distance and lane query similarity, named TopoLogic. This method mitigates the impact of endpoint shifts in geometric space, and introduces explicit similarity calculation in semantic space as a complement. By integrating results from both spaces, our method provides more comprehensive information for lane topology. Ultimately, our approach significantly outperforms the existing state-of-the-art methods on the mainstream benchmark OpenLane-V2 (23.9 v.s. 10.9 in TOP_{ll} and 44.1 v.s. 39.8 in OLS on *subset_A*). Additionally, our proposed geometric distance topology reasoning method can be incorporated into well-trained models without re-training, significantly boosting the performance of lane topology reasoning. The code is released at https://github.com/Franpin/TopoLogic.

*Corresponding Author

38th Conference on Neural Information Processing Systems (NeurIPS 2024).

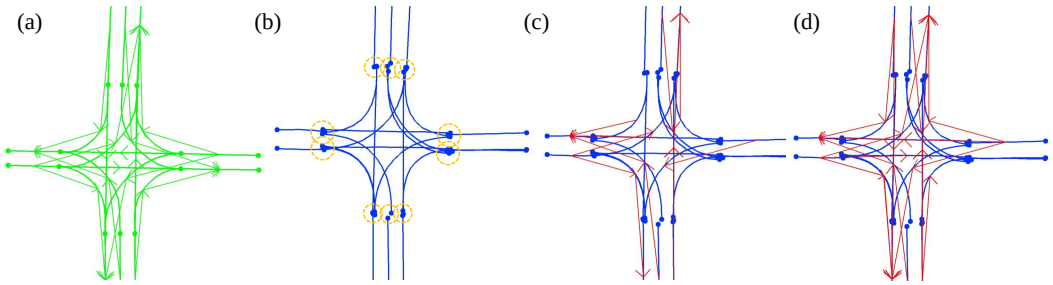


Figure 1: **Comparison of results with and without post-processing in TopoNet.** We use a post-processing based on geometric distance to improve the lane topology reasoning performance of TopoNet. **(a)** denotes the ground truth of lane topology reasoning. **(b)** denotes the endpoints of two connected lanes in prediction do not overlap (marked with yellow circle) as desired in ground truth. **(c)** denotes the lane topology reasoning result of TopoNet, the arrow denotes lane topology (marked with red arrow). **(d)** denotes the lane topology reasoning result of TopoNet using post-processing, significantly improves the reasoning precision of lane topology.

1 Introduction

In recent years, the field of autonomous driving has witnessed numerous milestone achievements and has progressively shifted from pure research to practical applications. In complex driving scenarios, vehicles need to perceive lanes & traffic elements and reason their topological relationships (i.e., lane connectivity and correspondence with traffic elements), which provides comprehensive information for downstream path planning and motion control. Under the trend of end-to-end autonomous driving, abovementioned perception and reasoning are integrated into a single task, referred to as topology reasoning [1] in autonomous driving scenes. This challenge has attracted widespread attention within the ego planning [2, 3, 4]and high-definition map learning [5, 6, 7, 8] communities.

The topology reasoning task has garnered significant attention recently, since it is closer to the real needs. Some works have explored lane centerline representation [1, 9, 10, 11] and lane segment representation [12], while others have introduced SDMap (Standard-Definition Map) [13] to provide additional clues for learning. However, existing works primarily focus on enhancements in the perception part, with scarce modifications made to the reasoning part. Irrespective of the approach details, existing studies typical employ vanilla MLP to learn lane topology directly from lane query. This paradigm has its shortcoming: since each lane is encoded independently through distinct query, it is challenging to ensure strictly overlap at the endpoints of two connected lanes, as shown in Figure 1(b). In contrast, it is evident that the endpoints of two connected lanes in the ground truth actually overlap perfectly, as shown in Figure 1(a). Lanes with slightly shifted endpoints may be erroneously classified by MLP as disconnected. This leads to MLP easily predicting fewer lane topology, as shown in Figure 1(c).

To tackle the aforementioned issues, we introduce TopoLogic, an interpretable method for lane topology reasoning that is based on lane geometric distances and the similarity of lane query in semantic space. The geometric distance-based approach aims to mitigates the impact of endpoint shift, thereby more robustly learning lane topology. This approach first calculates the geometric distance between lanes and then uses a learnable mapping function to map the distance to connectivity probability. Notably, for two given lanes, their geometric distance is defined as the distance between the ending point of one lane and the starting point of subsequent lane. This distance itself can serve as a strong criterion: when this distance is within a certain range, the predicted endpoints should be considered overlapping, and the lanes connected; otherwise, they are not. In this way, the lane topology reasoning becomes more tolerant of endpoint shifts, thus becoming more accurate. It's worth noting that even when the geometric distance method is merely applied as post-processing without re-training, the performance of SOTA model in lane topology reasoning is significantly improved, as shown in Figure 1(d). Moreover, reasoning lane topology completely based on geometric distances can lead to inaccuracies when lane detection is imprecise as is shown in Figure 4, since the calculation of lane geometric distance heavily relies on the accuracy of lane detection. To make up for the deficiency of geometric distances approach, we design a extra topology approach based on the similarity of lane

queries as a complement. This approach projects lane queries into a high-dimensional semantic space, and involves explicitly computing the dot product between lane query to determine similarity, and then mapping this similarity onto lane topology using sigmoid [14]. The approach for calculating lane query similarity complements the approach used for computing lane geometric distance topology and similarly boasts high interpretability. The final lane topology is obtained by fusing the topology matrix derived from both approachs. By the way, the lane topology is also used in GNN to enhance lane learning through the aggregation of features from adjacent lanes. **In summary, our contributions are as follows:**

1. We identify the current state of research in topology reasoning as "perception over reasoning", and reveal that the lane topology is easily disturbed by the endpoint shifts in lane detection when MLP employed solely for lane topology reasoning.

2. We propose an interpretable method, referred to as TopoLogic, which conducts lane topology reasoning by calculating lane geometric distances and semantic similarity of lane query in a high-dimensional semantic space.

3. Extensive experiments on the mainstream benchmark OpenLane-V2 for topology reasoning task indicate that our method significantly outperforms existing state-of-the-art methods, especially in lane topology metric. Even if employed solely as a post-processing step without re-training, proposed geometric distance approach can significantly enhance well-trained lane topology reasoning models.

2 Related Work

2.1 Lane Detection

Lane detection plays an important role in autonomous driving, which has been a fundamental aspect of lane topology reasoning. In the realm of lane detection, some works [15, 16, 17] attempt to perform lane detection on a segmentation map. Moreover, some researchers use vector-based methods to perform 3D lane detection [18, 19, 20, 21], however, these methods rely on a predetermined series of Y-axis coordinates within the query for forecasting 3D lanes, thereby lacking the capability to exclusively predict 3D lane positions along the Y-axis. In recent study, TopoNet [1] leverages Graph Neural Network (GNN) [22] to enhance the perception of lane centerline, while TopoMLP [9] utilizes PETR [23] for centerline detection. LaneSegNet [12] designs a Lane Attention mechanism to reinforce the perception of lane segment, and SMERF [13] introduces Standard-definition (SD) Map as an additional input to bolster the perception of lane centerline. In our work, we enhance lane learning by aggregating features of adjacent lanes through GNN, which involves computing lane geometric distance and lane query similarity.

2.2 Lane Topology Reasoning

In lane topology reasoning, accurate comprehension of lane topology is imperative for effective navigation and decision-making in autonomous driving. Some methods [24, 25, 26, 27, 28] have been proposed to address this. The STSU [29] model drew inspiration from DETR [30] and employed a neural network architecture, complemented by a multi-layer perceptron (MLP) to establish line connectivity. Building upon this foundation, Can et al. [31] introduced minimal cycle queries to refine centerline, ensuring accurate ordering of overlapping lines and thereby enhancing precision. Further advancements include the perception of centerline [1, 13, 11, 9] and the perception of lane segment [12]. Among them, CenterLineDet [11] and TopoNet[1], Both treat lane line as vertices and leverage an graph-based model to update lane representation and lane topology. While these methods have predominantly relied on MLP for generating adjacency matrices to represent lane topology. In our work, we calculate the lane topology matrices based on the geometric distances between lanes and the similarity of lane query within high-dimensional semantic space, respectively, and then fuse them to form the final lane topology. The fusion of geometric and semantic space enriches the model's understanding of lane topology, thereby culminating in improved performance in driving scene analysis and decision-making.

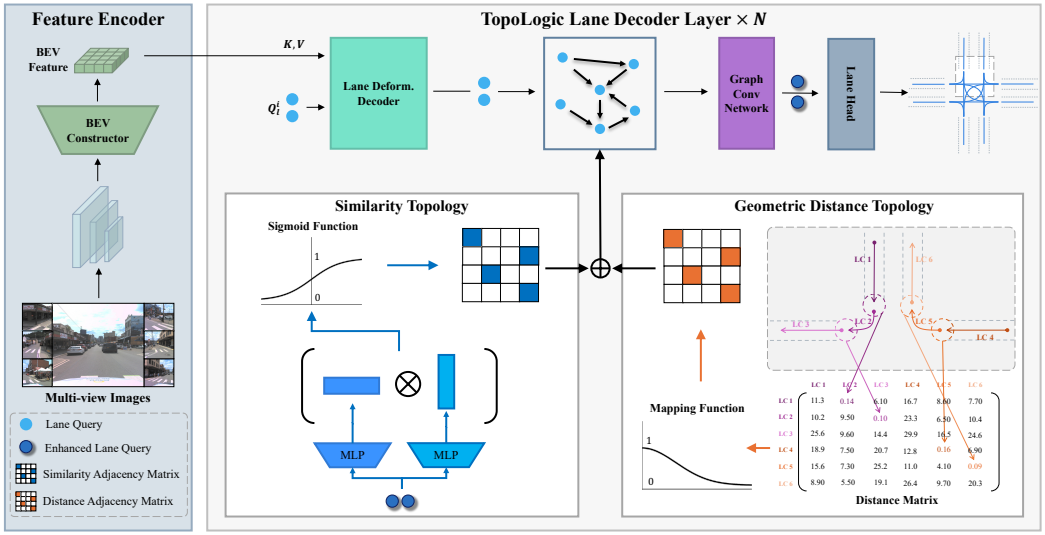


Figure 2: **Pipeline of TopoLogic.** The overarching structure of TopoLogic comprises two main components: an image encoder for feature extraction and transformation, and a lane decoder responsible for end-to-end topology reasoning. This decoder utilizes the proposed lane geometric distance topology and lane similarity topology, and fuse them into the final lane topology, which is facilitated through GNN to augment lane learning in the next decoder layer.

3 Method

3.1 Problem Definition

Given images captured by the surround-view cameras of a vehicle, lane topology reasoning needs to perceive lane instances in BEV(Bird's Eye View) and then infer the topology between these lane instances. The enhancement of lane instance perception assists in the reasoning of lane topology. Lane instances are described as a set of directed lane lines which is denoted as $L = [l_0, \ldots, l_{n-1}]$. Each lane line is composed of a series of ordered points, and it is denoted as $l = [p_0, \ldots, p_{n-1}], p = (x, y, z) \in \mathbb{R}^3$. The topology between lane instances signifies the connectivity of the directed lanes and it is depicted as a topology graph (V, E) , where the edge set $E \subseteq V \times V$. An entry (i, j) in E is positive if and only if the ending point of lane l_i^{end} connects to the starting point of lane l_j^{start}.

3.2 Overview

As is depicted in the Figure 2, our proposed TopoLogic takes multi-view images from onboard cameras as input. These images are processed by a backbone to generate multi-scale image features. Multi-scale image features are transformed into BEV features through a view transformation module, and then passed to a lane deformable decoder to generate lane query Q_l for lane detection. The proposed lane geometric distance approach and the lane similarity approach compute the lane topology respectively. Ultimately, the two topologies are fused and fed into GNN to augment the learning of lane line in the next decoder layer.

3.3 Lane Geometric Distance Topology

Lane Geometric Distance Matrix. Lane query Q_l can generate multiple directed lane lines through lane head. We can assess the connectivity between these lanes by computing the geometric distance between the ending point of one directed lane line and the starting point of the following lane line.

$$l_0, \ldots, l_{n-1} = \text{LaneHead}(Q_l^i) \tag{1}$$

$$d_{ij} = \mid l_i^{end} - l_j^{start} \mid \tag{2}$$

$$D = \{d_{ij} \mid i, j = 0 \ldots n-1\} \tag{3}$$

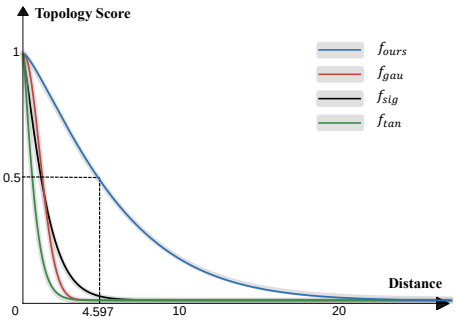


Figure 3: **Comparison of various mapping functions.** f_{gau} represent Gaussian function, f_{sig} represent sigmoid function, and f_{tan} represent tanh function. Compared to $f_{gau}, f_{sig}, f_{tan}$, our proposed function f_{ours} has greater tolerance for endpoint shift.

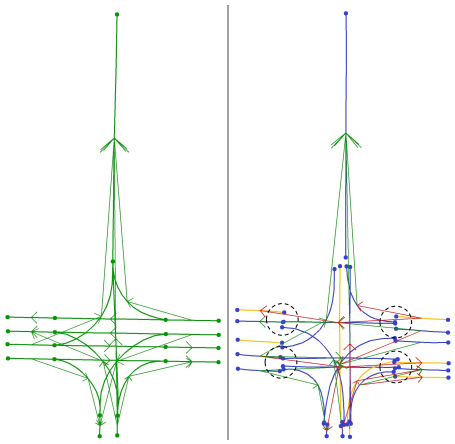


Figure 4: **Influence of inaccuracies in lane detection.** Blue denotes correct prediction of lane line, yellow denotes incorrect prediction of lane line, green denotes correct prediction of lane topology, and red denotes incorrect prediction of lane topology.

where D is the lane geometric distance, l_i and l_j represent preceding and subsequent lane lines, d_{ij} denotes the geometric distance between l_i and l_j, l_i^{end} signifies the last point of lane line l_i, and l_j^{start} indicates the first point of lane line l_j.

Distance to Topology Mapping Function. After obtaining the geometric distance matrix D for the lanes, it is necessary to map the lane geometric distance into the lane topology. The lane topology can be represented by a matrix ranged in 0∼1. Zero indicates that there is no connection between two lanes, while one indicates that there is a connection. This mapping function needs to capture the following notion: when the input $x \to 0$, it meaning the two lanes are very close to each other, the output $y \to 1$, suggesting that these two lanes are very likely to be connected. Conversely, as $x \to \infty$, $y \to 0$. Inspired by the Gaussian function, we design a learnable mapping function as follows:

$$f_{ours} = e^{-\frac{x^{\alpha}}{\lambda \cdot \sigma}} \tag{4}$$

where $x = d_{ij}$. σ is the standard deviation of the geometric distance matrix D. α, λ are learnable parameters. With the help of such mapping, we can get a lane topology as follows:

$$G_{dis} = \{f_{ours}(d_{ij}) | i, j = 0...n-1\} \tag{5}$$

There also exists some common alternative functions that meet the criteria, for example Gaussian function, sigmoid-based function and tanh-based function as Equation 6(a,b,c). We make a comparison between them with f_{ours} in Figure 3. Obviously, f_{ours} sets a larger geometric distance threshold for determining topological connectivity compared to f_{gau}, f_{sig} and f_{tan}, which makes the lane topology more robust to the endpoint shifts. Ablation study in Table 3 also verifies this opinion.

$$f_{gau} = e^{-\frac{x^2}{2}} \quad (a), \quad f_{sig} = \frac{2}{1+e^x} \quad (b), \quad f_{tan} = \frac{e^{-x} - e^x}{e^{-x} + e^x} + 1 \quad (c) \tag{6}$$

3.4 Lane Similarity Topology

Lane topology reasoning based on the geometric distance of lane lines can achieve commendable results when the detection of lane lines is accurate. However, since this topology reasoning method heavily relies on the detected lane lines, inaccuracies in lane line detection can interfere with the geometric approach and lead to erroneous reasoning outcomes, as demonstrated in the Figure 4. In light of this situation, we reason lane topology by calculating the similarity between lane query Q_l within high-dimensional semantic space. A higher similarity between Q_l indicates a greater likelihood of connectivity between the lanes, while lower similarity suggests an absence of connectivity. We

initially encode Q_l using two distinct MLP, and then represent the similarity by computing the inner product between the two encoding results. Finally, we require a function to map the similarity between Q_l to lane topology. Given the correlation between lane similarity and lane topology, we employ sigmoid to map the lane similarity onto lane topology. This process is as follows:

$$Q_{emb_1}, Q_{emb_2} = \mathrm{MLP}_1(Q_l^i), \mathrm{MLP}_2(Q_l^i) \tag{7}$$

$$S = \mathrm{matmul}(Q_{emb_1}, \mathrm{transpose}(Q_{emb_2})) \tag{8}$$

$$G_{sim} = \mathrm{sigmoid}(S) \tag{9}$$

where $Q_l^i \in \mathbb{R}^{N_l \times C}$, S represents the similarity of Q_l. $G_{sim} \in \mathbb{R}^{N_l \times N_l}$, while N_l represents the number of lane query.

3.5 Lane-Lane Topology

Both the lane topology reasoned from the geometric distance between lane lines and the lane topology reasoned from the similarity of lane query Q_l in the high-dimensional semantic space can indicate the connectivity of lanes. These two methods are complementary in the task of lane topology reasoning. In this context, we merge the two lane topology reasoning results together as the final and more accurate lane topology using learnable coefficients as follow:

$$G = \lambda_1 \cdot G_{dis} + \lambda_2 \cdot G_{sim} \tag{10}$$

where λ_1, λ_2 are learnable parameters, and G is the final lane topology prediction.

3.6 Learning

Similar to Transfomer-based networks [30, 32], the supervision is applied on each decoder layer to optimize the query feature iteratively. The overall loss of TopoLogic is:

$$\mathcal{L} = \mathcal{L}_{det} + \mathcal{L}_{top} \tag{11}$$

The lane detection loss $\mathcal{L}_{det}$ consists of a focal loss [33] for lane classification and an L1 loss for lane regression. The lane topology reasoning loss $\mathcal{L}_{top}$ includes only the loss computed for G_{sim}. As for the calculation of G_{dis}, since it is enhanced by GNN to facilitate lane learning, we update its learnable parameters through $\mathcal{L}_{det}$.

4 Experiment

4.1 Dataset and Metric

Dataset. We have evaluated TopoLogic on the OpenLane-V2 [34] , which is currently the only large-scale perception and topology reasoning dataset devised for autonomous driving scenarios. This dataset was developed by Argogorse2 [35] and nuScenes [36] respectively. It provides annotations for both lane centerline tasks and lane segment detection tasks. OpenLane-V2 consists of two subsets: *subset_A* and *subset_B*, each comprising 1000 scenes with 2Hz multi-view images and annotations. Each subset includes annotations for the lane centerline, traffic element, lane topology, as well as the topology between traffic element and lane. In *subset_A*, there are seven views as input, together with an additional Standard-definition Map input, and the annotations for lane segment have been expanded; *subset_B* contain only six views as input.

Metric. OpenLane-V2 evaluates perception tasks for both lane centerline and lane segment.

(1) In the task of lane centerline perception, the metrics include DET_l, DET_t, TOP_{ll}, and TOP_{lt}. DET_l quantifies similarity by averaging the Frechet distance at matching thresholds of 1.0, 2.0, 3.0. DET_t uses Intersection over Union (IoU) as a measure of similarity and calculates the average across different traffic categories. TOP_{ll} and TOP_{lt} respectively compute the topology matrix similarity between lanes and between lanes and traffic elements, with the overall evaluation metric for lane centerlines being denoted as OLS. The OLS is carried out as $\mathrm{OLS} = \frac{1}{4}[\mathrm{DET}_l + \mathrm{DET}_t + f(\mathrm{TOP}_{ll}) + f(\mathrm{TOP}_{lt})]$, where f is the square root function.

(2) In the task of lane segment perception, we adopt the metric proposed by LaneSegNet for evaluating lane segment perception. These include the lane segment distance D_{ls}, the corresponding Average

Precision AP_{ls} and AP_{ped}, with the mAP calculated as the average of AP_{ls} and AP_{ped}. The lane segment topology metric is denoted as TOP_{lsls}.

(3) For centerline, OpenLane-V2 has two versions available for evaluation on TOP_{ll}, TOP_{lt}, and OLS: v1.0.0 and v2.1.0. For lane segment, OpenLane-V2 has versions v2.0.0 and v2.1.0 on TOP_{lsls}. [2] **Since the ultimate goal of perception is reasoning, we believe that topology metrics are what should be paid more attention. Moreover, our modifications mainly involve lane topology reasoning, so we primarily focus on the lane topology metric TOP_{ll} and TOP_{lsls}.**

4.2 Implementation Details

Feature Extractor. All images are resized into the same resolution of 1550 × 2048. For reproducibility, we utilize the official implementations of TopoNet, SMERF, and LaneSegNet models. Both models use ResNet-50 [37] backbone pretrained on ImageNet [38] paired with a Feature Pyramid Network (FPN) [39] to extract multi-scale features. The number of output channels is set to 256. We employ the view transformer from BEVformer [40] encoder to transform multi-scale features into BEV features. The size of BEV grids is set to 200×100. TopoLogic is configured identically.

Lane Detector. We employ Deformable DETR [32] for the detection of lane line. The number of query is set to 200. After passing through each decoder layer of Deformable DETR, the query undergo GNN using the lane topology matrix. We predict the offset of the lane lines by setting reference points, with each lane line consisting of 11 three-dimensional points. For LaneHead, the classification head adopts a three-layer MLP with LayerNorm and ReLU to predict the confidence score of the lane line. The regression head is a three-layer MLP combined with ReLU, used to predict the 11×3 offset of the lane line. For lane detection loss $\mathcal{L}_{det_l}$, the weight of the classification part is 1.5, and the weight of the regression part is 0.025.

Lane Topology Head. The Lane Topology Head consists of a lane geometry distance predictor and a lane similarity predictor. For lane geometry distance predictor, we first calculate the geometric distance between the terminating point of the previous lane line and the starting point of the subsequent lane line to obtain a 200×200 distance matrix. Then the distance matrix is mapped to a lane topology matrix through a learnable mapping function$f(x) = e^{-\frac{x^\alpha}{\lambda \cdot \sigma}}$, where the size of α, λ, σ is 1×1, σ is the standard deviation of x, α, λ are learnable parameters, α is initialized to 2, λ is initialized to 0.2. For the calculation of lane similarity, given a 200×256 lane query, it is encoded through two different three-layer MLP. The we compute the similarity between the encoded results, resulting in a 200×200 lane similarity matrix. The similarity matrix is transformed into a lane topology matrix through sigmoid. The two lane topology matrices are fused into the final lane topology using learnable coefficients, which are initialized to 1 and have a size of 1×1.

Training. We train TopoLogic utilizing the AdamW optimizer [43] with a weight decay of 0.01 with an initial learning rate of 2×10^{-4} and employ a cosine annealing schedule for the learning rate. All experiment is trained for 24 epochs on 8 NVIDIA RTX 3090 GPUs with a batch size of 16.

4.3 Comparison to State-of-the-art

Centerline. We compared TopoLogic with existing state-of-the-art methods such as STSU, VectorMapNet, MapTR, TopoNet, SMERF on centerline. Table 1 shows the results on the *subset_A* and *subset_B* datasets. Without any additions, our method achieves state-of-the-art performance. Compared with TopoNet, our method achieves decent detection accuracy (**29.9** v.s. 28.6 on *subset_A*, **25.9** v.s. 24.4 on *subset_B*), especially with a significant improvement in TOP_{ll} (**18.6** v.s. 4.1 on *subset_A* for v1.0.0, **23.9** v.s. 10.8 on *subset_A* for v2.1.0), which is the lane topology reasoning score. There is also an improvement in OpenLane-V2 overall score OLS (**41.6** v.s. 35.6 on *subset_A* for v1.0.0, **44.1** v.s. 39.8 on *subset_A* for v2.1.0). Even when using SDMap, our proposed TopoLogic still manages to achieve state-of-the-art performance and realizes a significant improvement in TOP_{ll} (**23.4** v.s. 7.5 for v1.0.0, **28.9** v.s. 15.4 for v2.1.0).

Lane Segment. Concurrently, we compared TopoLogic with existing state-of-the-art methods such as TopoNet, MapTR, MapTRv2, LaneSegNet on lane segment. In Table 2, it indicates that our method exhibits improvements in the Mean Average Precision (mAP) for lane segment detection compared

[2] See the official repository for version differences of metrics: `https://github.com/OpenDriveLab/OpenLane-V2/issues/76d`

Table 1: Performance comparison with state-of-the-art methods on OpenLane-V2 benchmark on centerline. Results for existing methods are from TopoNet, TopoMLP and SMERF. "SDMap" indicates the use of a Standard-definition Map. "-" denotes the absence of relevant data. We are more focused on TOP_{ll}.

Data	Method	SDMap	DET_l ↑	DET_t ↑	v1.0.0			v2.1.0		
					TOP_{ll} ↑	TOP_{lt} ↑	OLS↑	TOP_{ll} ↑	TOP_{lt} ↑	OLS↑
subset_A	STSU [16]	×	12.7	43.0	0.5	15.1	25.4	2.9	19.8	29.3
	VectorMapNet [6]	×	11.1	41.7	0.4	5.9	20.8	2.7	9.2	24.9
	MapTR [41]	×	17.7	43.5	1.1	10.4	26.0	5.9	15.1	31.0
	TopoNet [1]	×	28.6	48.6	4.1	20.8	35.6	10.9	23.8	39.8
	TopoMLP [9]	×	28.3	**50.0**	7.2	22.8	38.2	19.0	23.4	42.2
	TopoLogic	×	**29.9**	47.2	**18.6**	**21.5**	**41.6**	**23.9**	**25.4**	**44.1**
	SMERF [13]	✓	33.4	**48.6**	7.5	23.4	39.4	15.4	25.4	42.9
	TopoLogic	✓	**34.4**	48.3	**23.4**	**24.4**	**45.1**	**28.9**	**28.7**	**47.5**
subset_B	STSU [16]	×	8.2	43.9	0.0	9.4	21.2	-	-	-
	VectorMapNet [6]	×	3.5	49.1	0.0	1.4	16.3	-	-	-
	MapTR [41]	×	15.2	54.0	0.5	6.1	25.2	-	-	-
	TopoNet [1]	×	24.3	55.0	2.5	14.2	33.2	6.7	16.7	36.8
	TopoMLP [9]	×	**26.6**	**58.3**	7.6	**17.8**	38.7	-	-	-
	TopoLogic	×	25.9	54.7	**15.1**	15.1	**39.6**	**21.6**	**17.9**	**42.3**

Table 2: Performance comparison with state-of-the-art methods on OpenLane-V2 benchmark on lane segment. Results for existing methods are from LaneSegNet. "-" denotes the absence of relevant data. We are more focused on TOP_{lsls}.

Method	mAP↑	AP_{ls} ↑	AP_{ped} ↑	v2.0.0	v2.1.0
				TOP_{lsls} ↑	TOP_{lsls} ↑
TopoNet [1]	23.0	23.9	22.0	1.0	-
MapTR [41]	27.0	25.9	28.1	-	-
MapTRv2 [42]	28.5	26.6	30.4	-	-
LaneSegNet [12]	32.6	32.3	32.9	8.1	25.4
TopoLogic	**33.2**	**33.0**	**33.4**	**22.0**	**30.8**

with LaneSegNet (**33.2** v.s. 32.6). Moreover, there is a significant enhancement in topology reasoning score TOP_{lsls} (**22.0** v.s. 8.1 on v2.0.0, **30.8** v.s. 25.4 on v2.1.0).

4.4 Alation Study

We have studied several important components of TopoLogic and conducted ablation experiments on the OpenLane-V2 *subset_A*. In the following text, we employ evaluation metrics from the latest v2.1.0 release for our assessment.

The design of mapping function. We have studied the effect of different mapping functions in the transformation from lane geometric distances to lane topology. Table 3 suggests that our designed learnable mapping function performs better at mapping the geometric distances of lane lines to lane topology compared to sigmoid-based, tanh-based, and Gaussian functions. It exhibits the best performance in terms of both the lane topology reasoning score and the centerline score (**29.9** v.s. 28.9 v.s. 28.7 v.s. 27.6 on DET_l, and **23.9** v.s. 21.7 v.s. 19.1 v.s. 15.1 on TOP_{ll}).

The approach of lane topology reasoning. We have investigated the impact of various lane topology computation approaches on lane topology reasoning, specifically using MLP, lane query similarity, and geometric distance. The results in the Table 4 indicate that the integration of a lane topology method, combining both lane geometric distance and lane query similarity calculations, yields optimal results on TOP_{ll} (**23.9** v.s. 20.1 v.s. 12.9 v.s. 10.8) and also performs best in terms of lane line detection scores indicated by DET_l (**29.9** v.s. 28.6 v.s. 28.1 v.s. 27.8). This demonstrates that

Table 3: Ablation study on different mapping functions from lane geometric distance to lane topology on centerline.

Function	DET_l ↑	DET_t ↑	TOP_{ll} ↑	TOP_{lt} ↑	OLS↑
f_{tan}	27.6	47.2	15.1	24.8	40.9
f_{sig}	28.7	44.1	19.1	24.1	41.4
f_{gau}	28.9	46.8	21.7	23.2	42.6
$\boldsymbol{f}_{ours}$	**29.9**	**47.2**	**23.9**	**25.4**	**44.1**

Table 4: Ablation study on different lane topology reasoning approaches on centerline. Ours indicate Similarity+GeoDist.

Approach	DET_l ↑	DET_t ↑	TOP_{ll} ↑	TOP_{lt} ↑	OLS↑
MLP	27.8	46.8	10.8	23.9	39.1
Similarity	28.1	46.4	12.9	23.7	39.8
GeoDist	28.6	44.1	20.1	23.1	41.4
Ours	**29.9**	**47.2**	**23.9**	**25.4**	**44.1**

Table 5: Ablation study on using MLP to encode lane query computing lane similarity topology. Ours indicate using two indepent MLPs.

Approach	DET_l ↑	DET_t ↑	TOP_{ll} ↑	TOP_{lt} ↑	OLS↑
No MLP	25.6	46.5	18.7	20.8	40.2
Single MLP	27.5	46.8	21.2	23.8	42.3
Ours	**29.9**	**47.2**	**23.9**	**25.4**	**44.1**

the topology obtained from the fusion of these two methods can also improve the learning of lane centerline through GNN feature enhancement.

The approach of similarity topology. We investigated the impact of using MLPs to encode lane queries for computing lane similarity on lane topology reasoning. As Lane similarity topology, its subtlety lies in that it uses two independent MLPs to map the lane query rather than a single MLP, which can decouple a lane into two queries of start and end point, achieving an analogous effect to the geometric distance approach in semantical space. The results presented in Table 5 indicate that encoding lane query with two independent MLPs to obtain features for the starting and ending points of lanes allows for better computation of lane similarity topology. This approach exhibits the best performance in terms of both lane centerline detection score and lane topology reasoning score (**29.9** v.s. 27.5 v.s. 25.6 on DET_l, and **23.9** v.s. 21.2 v.s. 18.7 on TOP_{ll}).

The post-processing mode of geometric distance approach. We have investigated the effectiveness of geometric distance approach as a post-processing module on well-trained model. In Table 6, we conducted experiments by adding a post-processing to the already trained TopoNet, SMERF, and LaneSegNet, respectively. The results indicate that our proposed approach, which calculates lane topology based on lane geometric distances, can be integrated into well-trained models without any additional modifications and can significantly enhance the performance of lane topology reasoning(**22.3** v.s. 10.9 on TopoNet, **26.2** v.s. 15.4 v.s. on SMERF).

4.5 Qualitative Analysis

As shown in Figure 5, we presents a qualitative comparison between TopoLogic and TopoNet. Specifically, two traffic scenes are selected for analysis, and the results of lane line detection and topology reasoning are visualized. The first row displays multi-view inputs of realistic scenes, while the second row shows the lane detection results of TopoLogic and TopoNet alongside the ground truth. Notably, TopoLogic demonstrates superior accuracy in lane line detection compared to TopoNet.

Lane Graph. The inherent complexity of topology reasoning makes intuitive representation of results challenging. To address this issue, as shown in third row in Figure 5, we construct a lane graph where nodes represent lane lines, and their relative positions in the graph correspond one-to-one with the relative positions of lane lines. This layout enhances the connection between lane line detection results and facilitates subsequent analysis. Additionally, we use directed edges to represent the lane topology, with red indicating error predictions and blue indicating missing predictions. TopoLogic consistently exhibits proficient lane topology reasoning across various intersection scenarios, showcasing significantly enhanced performance in the topology graph compared to TopoNet.

Table 6: Ablation study on incorporating lane geometric distance into post-processing for well-trained model under different task settting (centerline / centerline+SDMap / lane segment).

Centerline	TOP_{ll} ↑	Centerline+SDMap	TOP_{ll} ↑	Lane Segment	TOP_{lsls} ↑
TopoNet [1]	10.9	SMERF [13]	15.4	LaneSegNet [12]	25.4
TopoNet+GeoDist	**22.3**	**SMERF+GeoDist**	**26.2**	**LaneSegNet+GeoDist**	**29.6**

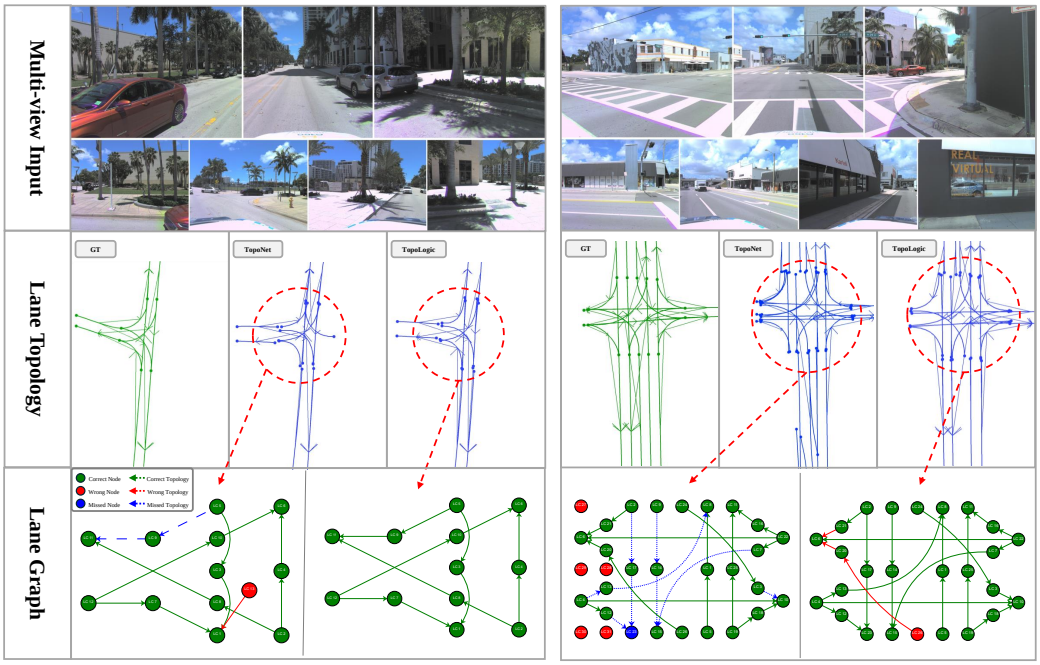


Figure 5: **Qualitative result about lane topology reasoning result of TopoNet and our TopoLogic.** The first row denotes multi-view inputs. The second row denotes lane detection result and lane topology reasoning result. The third row denotes graph form of lane topology reasoning (node indicates lane line, edge indicates lane topology), where green color indicates the right prediction, while red color indicates the error prediction and blue color indicates missing prediction.

5 Conclusion

In this paper, we reveal the limitation of using vanilla MLP in lane topology reasoning task and propose TopoLogic, which is the first to employ an interpretable approach for lane topology reasoning. TopoLogic fuses the geometric distance of lane line endpoints mapped through a designed function and the similarity of lane query in a high-dimensional semantic space to reason lane topology. Experiments on the large-scale autonomous driving dataset OpenLane-V2 benchmark demonstrate that TopoLogic significantly outperforms existing methods in topology reasoning in complex scenarios.

Limitations. Due to the GNN's role in merely aggregating features from adjacent lanes to enhance the learning of the current lane, our proposed method significantly improves the performance of lane topology but does not substantially elevate lane detection.

Impact. Based on the previous sections, it is evident that our proposed method is intended for research purposes. It should not be directly used in or deployed within any actual autonomous driving application. Notably, we cannot provide any guarantee in safety-critical situations.

6 Acknowledgements

This work is supported by National Key R&D Program of China (2023YFD2000303) and National Natural Science Foundation of China (62372433).

References

[1] Tianyu Li, Li Chen, Huijie Wang, Yang Li, Jiazhi Yang, Xiangwei Geng, Shengyin Jiang, Yuting Wang, Hang Xu, Chunjing Xu, Junchi Yan, Ping Luo, and Hongyang Li. Graph-based topology reasoning for driving scenes, 2023.

[2] Yuning Chai, Benjamin Sapp, Mayank Bansal, and Dragomir Anguelov. Multipath: Multiple probabilistic anchor trajectory hypotheses for behavior prediction. In *CoRL*, 2020.

[3] Sergio Casas, Abbas Sadat, and Raquel Urtasun. Mp3: A unified model to map, perceive, predict and plan. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14403–14412, June 2021.

[4] Yihan Hu, Jiazhi Yang, Li Chen, Keyu Li, Chonghao Sima, Xizhou Zhu, Siqi Chai, Senyao Du, Tianwei Lin, Wenhai Wang, et al. Planning-oriented autonomous driving. In *CVPR*, 2023.

[5] Qi Li, Yue Wang, Yilun Wang, and Hang Zhao. Hdmapnet: An online hd map construction and evaluation framework. In *ICRA*, 2022.

[6] Yicheng Liu, Tianyuan Yuan, Yue Wang, Yilun Wang, and Hang Zhao. Vectormapnet: End-to-end vectorized hd map learning. In *ICML*, 2023.

[7] Limeng Qiao, Wenjie Ding, Xi Qiu, and Chi Zhang. End-to-end vectorized hd-map construction with piecewise bezier curve. In *CVPR*, 2023.

[8] Wenjie Ding, Limeng Qiao, Xi Qiu, and Chi Zhang. Pivotnet: Vectorized pivot learning for end-to-end hd map construction. In *ICCV*, 2023.

[9] Dongming Wu, Jiahao Chang, Fan Jia, Yingfei Liu, Tiancai Wang, and Jianbing Shen. Topomlp: An simple yet strong pipeline for driving topology reasoning. *ICLR*, 2024.

[10] Dongming Wu, Fan Jia, Jiahao Chang, Zhuoling Li, Jianjian Sun, Chunrui Han, Shuailin Li, Yingfei Liu, Zheng Ge, and Tiancai Wang. The 1st-place solution for cvpr 2023 openlane topology in autonomous driving challenge. *arXiv preprint arXiv:2306.09590*, 2023.

[11] Zhenhua Xu, Yuxuan Liu, Yuxiang Sun, Ming Liu, and Lujia Wang. Centerlinedet: Centerline graph detection for road lanes with vehicle-mounted sensors by transformer for hd map generation. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pages 3553–3559. IEEE, 2023.

[12] Tianyu Li, Peijin Jia, Bangjun Wang, Li Chen, Kun Jiang, Junchi Yan, and Hongyang Li. Lanesegnet: Map learning with lane segment perception for autonomous driving. In *ICLR*, 2024.

[13] Katie Z Luo, Xinshuo Weng, Yan Wang, Shuang Wu, Jie Li, Kilian Q Weinberger, Yue Wang, and Marco Pavone. Augmenting lane perception and topology understanding with standard definition navigation maps. *arXiv preprint arXiv:2311.04079*, 2023.

[14] Stefan Elfwing, Eiji Uchibe, and Kenji Doya. Sigmoid-weighted linear units for neural network function approximation in reinforcement learning, 2017.

[15] Anil Batra, Suriya Singh, Guan Pang, Saikat Basu, CV Jawahar, and Manohar Paluri. Improved road connectivity by joint learning of orientation and segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10385–10393, 2019.

[16] Yigit Baran Can, Alexander Liniger, Danda Pani Paudel, and Luc Van Gool. Topology preserving local road network estimation from single onboard camera image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17263–17272, 2022.

[17] Songtao He and Hari Balakrishnan. Lane-level street map extraction from aerial imagery. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 2080–2089, 2022.

[18] Noa Garnett, Rafi Cohen, Tomer Pe'er, Roee Lahav, and Dan Levi. 3d-lanenet: end-to-end 3d multiple lane detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 2921–2930, 2019.

[19] Yuliang Guo, Guang Chen, Peitao Zhao, Weide Zhang, Jinghao Miao, Jingao Wang, and Tae Eun Choe. Gen-lanenet: A generalized and scalable approach for 3d lane detection. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXI 16*, pages 666–681. Springer, 2020.

[20] Fan Yan, Ming Nie, Xinyue Cai, Jianhua Han, Hang Xu, Zhen Yang, Chaoqiang Ye, Yanwei Fu, Michael Bi Mi, and Li Zhang. Once-3dlanes: Building monocular 3d lane detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17143–17152, 2022.

[21] Li Chen, Chonghao Sima, Yang Li, Zehan Zheng, Jiajie Xu, Xiangwei Geng, Hongyang Li, Conghui He, Jianping Shi, Yu Qiao, et al. Persformer: 3d lane detection via perspective transformer and the openlane benchmark. In *European Conference on Computer Vision*, pages 550–567. Springer, 2022.

[22] Franco Scarselli, Marco Gori, Ah Chung Tsoi, Markus Hagenbuchner, and Gabriele Monfardini. The graph neural network model. *IEEE Transactions on Neural Networks*, 20(1):61–80, 2009.

[23] Yingfei Liu, Tiancai Wang, Xiangyu Zhang, and Jian Sun. Petr: Position embedding transformation for multi-view 3d object detection. In *ECCV*, 2022.

[24] Songtao He, Favyen Bastani, Satvat Jagwani, Mohammad Alizadeh, Hari Balakrishnan, Sanjay Chawla, Mohamed M Elshrif, Samuel Madden, and Mohammad Amin Sadeghi. Sat2graph: Road graph extraction through graph-tensor encoding. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXIV 16*, pages 51–67. Springer, 2020.

[25] Jannik Zürn, Johan Vertens, and Wolfram Burgard. Lane graph estimation for scene understanding in urban driving. *IEEE Robotics and Automation Letters*, 6(4):8615–8622, 2021.

[26] Songtao He and Hari Balakrishnan. Lane-level street map extraction from aerial imagery. In *2022 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 1496–1505, 2022.

[27] Wele Gedara Chaminda Bandara, Jeya Maria Jose Valanarasu, and Vishal M Patel. Spin road mapper: Extracting roads from aerial images via spatial and interaction space graph reasoning for autonomous driving. In *2022 International Conference on Robotics and Automation (ICRA)*, pages 343–350. IEEE, 2022.

[28] Namdar Homayounfar, Wei-Chiu Ma, Justin Liang, Xinyu Wu, Jack Fan, and Raquel Urtasun. Dagmapper: Learning to map by discovering lane topology. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 2911–2920, 2019.

[29] Yigit Baran Can, Alexander Liniger, Danda Pani Paudel, and Luc Van Gool. Structured bird's-eye-view traffic scene understanding from onboard images. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 15641–15650, 2021.

[30] Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. End-to-end object detection with transformers. page 213–229, Berlin, Heidelberg, 2020. Springer-Verlag.

[31] Yigit Baran Can, Alexander Liniger, Danda Pani Paudel, and Luc Van Gool. Topology preserving local road network estimation from single onboard camera image. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 17242–17251, 2022.

[32] Xizhou Zhu, Weijie Su, Lewei Lu, Bin Li, Xiaogang Wang, and Jifeng Dai. Deformable detr: Deformable transformers for end-to-end object detection. In *ICLR*, 2021.

[33] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *ICCV*, 2017.

[34] Huijie Wang, Tianyu Li, Yang Li, Li Chen, Chonghao Sima, Zhenbo Liu, Bangjun Wang, Peijin Jia, Yuting Wang, Shengyin Jiang, Feng Wen, Hang Xu, Ping Luo, Junchi Yan, Wei Zhang, and Hongyang Li. Openlane-v2: A topology reasoning benchmark for unified 3d hd mapping. In *NeurIPS*, 2023.

[35] Benjamin Wilson, William Qi, Tanmay Agarwal, John Lambert, Jagjeet Singh, Siddhesh Khandelwal, Bowen Pan, Ratnesh Kumar, Andrew Hartnett, Jhony Kaesemodel Pontes, et al. Argoverse 2: Next generation datasets for self-driving perception and forecasting. In *NeurIPS*, 2021.

[36] Holger Caesar, Varun Bankiti, Alex H Lang, Sourabh Vora, Venice Erin Liong, Qiang Xu, Anush Krishnan, Yu Pan, Giancarlo Baldan, and Oscar Beijbom. nuscenes: A multimodal dataset for autonomous driving. In *CVPR*, 2020.

[37] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, 2016.

[38] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C. Berg, and Li Fei-Fei. Imagenet large scale visual recognition challenge. 1409.0575, 2014.

[39] Yangyan Li, Sören Pirk, Hao Su, Charles Ruizhongtai Qi, and Leonidas J. Guibas. FPNN: field probing neural networks for 3d data. 2016.

[40] Zhiqi Li, Wenhai Wang, Hongyang Li, Enze Xie, Chonghao Sima, Tong Lu, Yu Qiao, and Jifeng Dai. Bevformer: Learning bird's-eye-view representation from multi-camera images via spatiotemporal transformers. In *ECCV*, 2022.

[41] Bencheng Liao, Shaoyu Chen, Xinggang Wang, Tianheng Cheng, Qian Zhang, Wenyu Liu, and Chang Huang. Maptr: Structured modeling and learning for online vectorized hd map construction. *arXiv preprint arXiv:2208.14437*, 2022.

[42] Bencheng Liao, Shaoyu Chen, Yunchi Zhang, Bo Jiang, Qian Zhang, Wenyu Liu, Chang Huang, and Xinggang Wang. MapTRv2: An end-to-end framework for online vectorized hd map construction. *arXiv preprint arXiv:2308.05736*, 2023.

[43] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. 2015.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: We have clearly stated this in the introduction.

   Guidelines:

   - The answer NA means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A No or NA answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: We have discussed this in the conclusion of the paper.

   Guidelines:

   - The answer NA means that the paper has no limitation while the answer No means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory Assumptions and Proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

Justification: We have provided this in the method section.

Guidelines:

- The answer NA means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental Result Reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: We have provided implementation detail in the experiment section.

   Guidelines:

   - The answer NA means that the paper does not include experiments.
   - If the paper includes experiments, a No answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
     (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
     (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
     (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
     (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: We have provided the data and code in supplemental material.

Guidelines:

- The answer NA means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://nips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so "No" is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://nips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental Setting/Details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

   Answer: [Yes]

   Justification: We have provided implementation detail in the experiment section.

   Guidelines:

   - The answer NA means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment Statistical Significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [No]

   Justification: Due to limitation in computational resource, we did not conduct multiple iterations of the same experiment to calculate error.

   Guidelines:

   - The answer NA means that the paper does not include experiments.
   - The authors should answer "Yes" if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).

- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g. negative error rates).
- If error bars are reported in tables or plots, The authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments Compute Resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: We have provided the computer resources necessary to reproduce the experiments in implementation detail of the experiment section.

   Guidelines:

   - The answer NA means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code Of Ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: We adhere to the NeurIPS Code of Ethics in every respect.

   Guidelines:

   - The answer NA means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer No, they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader Impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: We have mentioned the impact in the conclusion.

    Guidelines:

    - The answer NA means that there is no societal impact of the work performed.
    - If the authors answer NA or No, they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.

- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

    Answer: [NA]

    Justification: Our paper poses no such risks.

    Guidelines:

    - The answer NA means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: They are properly credited and properly respected.

    Guidelines:

    - The answer NA means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.

- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New Assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: New assets introduced in the paper well are documented and the documentation is provided alongside the assets.

    Guidelines:

    - The answer NA means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and Research with Human Subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [NA]

    Justification: The paper does not involve crowdsourcing nor research with human subjects.

    Guidelines:

    - The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [NA]

    Justification: The paper does not involve crowdsourcing nor research with human subjects.

    Guidelines:

    - The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.